Document 4 -- Additional Information

Aran Technologies | MBC-3 Registration | Max 300 words

Simulation Demonstration -- Phase I Ready

A verified five-drone swarm simulation demonstrates all core MBC-3 requirements on ROS2 Jazzy and Gazebo Harmonic: real-time ASP generation, bully-protocol leader election, graceful degradation on drone loss, and sector reassignment -- 344 radar tracks logged with four surviving drones sustaining full mission continuity.

Single-drone ISR video prepared for Phase I: 4-minute 1920x1080 recording covering survey grid (11 WPs), primary target orbit (50 m radius, locked +/-0.5 m), and RTL with 3D occupancy map save. Mission exit verified clean (v12-MPC-v5).

Indigenisation Breakdown

Sub-system	Indigenous Content
Hexacopter airframe	Indigenous design and local fabrication
Antenna panels (6 x per drone)	Indigenous PCB design
AI software stack (CFAR, RF, LLM)	100% indigenous (Python / C++, open-source models)
GCS dashboard (Flask / SocketIO)	100% indigenous
ROS2 swarm coordination stack	100% indigenous
Flight controller firmware	Custom STM32 PCB -- designed from scratch
Compute module (Jetson)	COTS -- imported
Radar frontend (AWR1843)	COTS -- imported
Mesh radio (Doodle Labs)	COTS -- imported

Estimated indigenisation: >= 55% by mission-criticality weighting -- meets MBC-3 §2.25 requirement.

Team

Aran Technologies is a three-member engineering startup building indigenous defence UAS systems from the ground up.

Kishore Udhayakumar | Founder & CEO -- Mechanical Engineering. Leads product development, mechanical design, thermal engineering, and business strategy. Drives go-to-market planning and direct engagement with Indian defence end-users.

Vegi Tejo Bhargav | Co-Founder & CTO -- Mechanical Engineering. Manages project execution, hexacopter airframe fabrication, additive manufacturing, procurement, and hardware integration roadmap.

L. Harsha Vardhan Naidu | Product Head, Electronics -- EEE. Designed the complete autonomous mission stack: PX4 SITL pipeline, ROS2 swarm middleware, GCS dashboard, STM32 flight controller firmware, and three-layer onboard AI pipeline (CFAR -> Random Forest -> LLM).

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Phase I Readiness

Simulation stack, GCS dashboard, and graceful degradation demonstration are prepared for live presentation at New Delhi, 13-24 July 2026.